

Which rewards produce behavioral contrast?

Current reward potency, neural reference memory, and behavioral expression are separable stages.

	Current U → behavior	History → VTA DA	History → behavior
Natural 100E/20E 11 mice	PASS Current reward quality changes persistence $\Delta \log\text{-duration}=+1.043$; 11/11; $P=.000977$	PASS Past reward history changes DA unique $\Delta R^2=.0269$; $P=.00670$	PASS Frozen reference predicts persistence $\beta=.476$, $P=.00566$; hazard OR=.459, $P=.000180$ <i>Reference is expressed in both DA and behavior.</i>
QE / quinine history 7 mice	PASS Current QE devaluation changes persistence $\Delta \log\text{-duration}=+2.013$ (EE–QE); 7/7; $P=.015625$	PASS Same current 100E, different history changes DA AUC $\beta=114.98$, $P=.00260$; circular-shift $P=5\times 10^{-5}$	PROVISIONAL Same current 100E: history predicts duration $\beta=.818$, cluster $P=.034$; serial-shift one-sided $P=.0287$ <i>Sensory/devaluation history reaches behavior, but n is small.</i>
VTA stimulation history 13 mice	PASS Acute ON stimulation changes persistence $\Delta \log\text{-duration}=+.208$; 11/13; $P=.00342$	PASS Past stimulation history changes later DA unique $\Delta R^2=.002524$; conditional $P=5\times 10^{-5}$	NULL Past stimulation history does not change duration $p=.742$ <i>Neural reference persists, but behavioral output gain is weak.</i>

Key dissociation: Stim13 shows that a reward can be acutely behaviorally potent and leave a persistent neural history trace, yet still fail to produce gross behavioral contrast.

QE behavioral-history evidence is labeled provisional because the same-current subset is 63 bouts from 7 mice; animal-level slopes are 5/6 positive but not significant (one-sided $P=.156$).